The **Horizon dashboard** serves as the official web-based graphical user interface (GUI) for the OpenStack project 1, 2. It provides a centralized, user-friendly environment for both administrators and end-users to interact with and manage their cloud resources 2.

Purpose of Horizon

The primary purpose of Horizon is to **simplify cloud management** by providing an intuitive interface that replaces or complements complex command-line interactions 3. It acts as a visual gateway to the various OpenStack services, enabling **self-service resource provisioning** 4, 5.

Features of the Dashboard Interface

Horizon offers a comprehensive suite of management tools:

- **Virtual Machine (VM) Management:** Users can launch, stop, reboot, and manage the lifecycle of VMs, including configuring hardware specifications (flavors) and security groups 1, 6, 7.
- **Networking:** The interface allows for the creation and configuration of virtual networks, subnets, and IP addresses 1, 7.
- **Image Management:** Users can upload, manage, and share VM images that serve as deployment templates 7.
- **Storage Access:** Provides management for both **block storage** (volumes for VMs) and **object storage** (unstructured data like logs or backups) 7.
- **Monitoring and Visualization:** It provides a visual overview of resource utilization—such as CPU, memory, and network traffic—to help identify bottlenecks 8.
- **Customization:** Users can personalize their dashboard layout, and administrators can extend the interface with custom panels or plugins 4, 8.

User Interaction Workflow

1. **Authentication:** Users log in, and Horizon interacts with the **Keystone Identity service** to verify credentials and assign appropriate roles 6, 9.
2. **Resource Selection:** Through the sidebar navigation, users select the resource type they need (e.g., "Instances" for VMs or "Networks") 7.
3. **Task Execution:** Instead of manual coding, users follow guided wizards to provision resources—for example, choosing an image, defining a network, and launching a VM in a single workflow 1, 3, 10.
4. **Feedback:** The dashboard provides immediate visual status updates on the success or failure of these operations 8.

Administration Capabilities

Cloud administrators use Horizon for high-level infrastructure oversight:

- **Identity Management:** Creating and managing user accounts and projects (isolated workspaces) 6.
- **Role-Based Access Control (RBAC):** Assigning specific permissions and roles to users within different projects 6.

- **Quota Management:** Setting limits on the number of resources (like VMs or vCPUs) a specific project can consume 5, 10.
- **System Health Monitoring:** Monitoring the overall health of the cloud infrastructure and managing resource availability 8, 10.

Advantages of Web-Based Cloud Management

- **Lower Learning Curve:** It is accessible to users without extensive command-line experience, promoting faster cloud adoption 3, 10.
- **Self-Service Environment:** Users can manage their own resources without waiting for administrative intervention, leading to faster application deployment 4.
- **Centralized Visibility:** Offers a "single pane of glass" view to monitor resource utilization and troubleshoot issues across the entire cloud 3.
- **Reduced Complexity:** Simplifies resource provisioning tasks that would otherwise require multiple complex API calls or CLI commands 3, 10.

Exam-Oriented Explanation

- **Definition:** Horizon is the official **OpenStack Web GUI** 2.
- **Key Concept:** It facilitates **Self-Service Cloud Management**, allowing users to provision their own compute, network, and storage resources 4.
- **Service Integration:** It acts as a frontend for core services: **Nova** (VMs), **Neutron** (Networks), **Cinder** (Storage), **Glance** (Images), and **Keystone** (Identity) 1, 6, 7.
- **User Roles:**
- **Admins:** Manage projects, users, quotas, and cloud health 10.
- **Developers/Users:** Provision and manage resources for their specific applications 5.
- **Deployment Value:** It reduces operational complexity and provides **centralized visibility** into the infrastructure's resource consumption 3, 10.